How-To Library

Informatica

Address Verification Best Practices for Singapore

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Abstract

This document tells you how to connect the elements in a Singapore address to the appropriate address elements in Informatica® Address Verification. This document also describes best practices that you can implement when you verify addresses in Singapore.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses in Singapore. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Singapore address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Singapore addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE
- PARSE

Character Set

Singapore reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Singapore addresses in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Singapore postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [BLOCK NUMBER, HOUSE NUMBER, OR BUILDING NUMBER] STREET
Line 3: [SUB-BUILDING INFORMATION][BUILDING INFORMATION]
Line 4: CITY POSTAL CODE
Line 5: COUNTRY
```

The following sample address illustrates the structure:

```
528 ANG MO KIO AVENUE 10
FLOOR 01 UNIT 2385 CHENG SAN CENTRE
SINGAPORE 560528
SGP
```

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name, Street Type	ANG MO KIO AVENUE 10	Street
House Number	528	HouseNumber
Postal Code	560528	PostalCode
City	SINGAPORE	Locality
Building Name	CHENG SAN CENTRE	Building
Floor	FLOOR 01	Sub-building 1
Apartment	UNIT 2385	Sub-building 2

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
  <AddressElements>
        <Country Item="1" Type="NAME">SINGAPORE</Country>
  </AddressElements>
  <AddressLines>
        <FormattedAddressLine Line="1">528 ANG MO KIO AVENUE 10</FormattedAddressLine>
```

```
        <FormattedAddressLine Line="2">FLOOR 01 UNIT 2385 CHENG SAN CENTRE</
FormattedAddressLine>
        <FormattedAddressLine Line="3">SINGAPORE 560528</FormattedAddressLine>
   </AddressLines>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.14.0.47331" -->

<Result     ProcessStatus="V4"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="SGP"
     PreferredScript="DATABASE"
     PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="F0F000F0F000F0FF00E0"
     ElementInputStatus="60600060600060660060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000000000000000"
     ElementRelevance="10100010100010110010"
     AddressType="B"
     LanguageISO3="ENG">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">SINGAPORE</Country>
        <Locality Item="1">SINGAPORE</Locality>
        <PostalCode Item="1">560528</PostalCode>
        <Street Item="1">ANG MO KIO AVENUE 10</Street>
        <Number Item="1">528</Number>
        <Building Item="1">CHENG SAN CENTRE</Building>
        <SubBuilding Item="1">FLOOR 01</SubBuilding>
        <SubBuilding Item="2">UNIT 2385</SubBuilding>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">528 ANG MO KIO AVENUE 10</DeliveryAddressLine>
        <DeliveryAddressLine Line="2">#01-2385 CHENG SAN CENTRE</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">SINGAPORE 560528</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">528 ANG MO KIO AVENUE 10</FormattedAddressLine>
        <FormattedAddressLine Line="2">#01-2385 CHENG SAN CENTRE</FormattedAddressLine>
        <FormattedAddressLine Line="3">SINGAPORE 560528</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>528 ANG MO KIO AVENUE 10
#01-2385 CHENG SAN CENTRE
SINGAPORE 560528</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Singapore address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
    <soap:Header/>
    <soap:Body>
        <v:Process>
            <v:login>****</v:login>
            <v:password>****</v:password>
            <v:parameters>
                <v:ProcessMode>BATCH</v:ProcessMode>
            <v:ServiceParameters></v:ServiceParameters>
```

```
            <v:ValidationParameters> </v:ValidationParameters>
            </v:parameters>
            <v:addresses>
                <v:Address>
                    <v:FormattedAddress>
                        <v:string>528 ANG MO KIO AVENUE 10</v:string>
                        <v:string>FLOOR 01 UNIT 2385 CHENG SAN CENTRE</v:string>
                        <v:string>SINGAPORE 56052</v:string>
                        <v:string>SINGAPORE</v:string>
                    </v:FormattedAddress>
                </v:Address>
            </v:addresses>
            <v:enrichments>
            </v:enrichments>
        </v:Process>
    </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <soap:Body>
        <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
            <ProcessResult>
                <StatusCode>100</StatusCode>
                <StatusMessage>OK</StatusMessage>
                <Results>
                    <Result>
                        <ProcessStatus>V4</ProcessStatus>
                        <CountryISO3>SGP</CountryISO3>
                        <ResultDataSet>
                            <ResultData>
                                <ResultNumber>1</ResultNumber>
                                <MailabilityScore>5</MailabilityScore>
                                <ResultPercentage>100.00</ResultPercentage>
                                <ElementInputStatus>6060006060006606006O</ElementInputStatus>
                                <ElementResultStatus>F0F000F0F000F0FF00E0</ElementResultStatus>
                                <ElementRelevance>10100010100010110010</ElementRelevance>
                                <ExtElementStatus>00000000000000000000</ExtElementStatus>
                                <AddressResolutionCode>0000000000000000000</
AddressResolutionCode>
                                <AddressType>B</AddressType>
                                <LanguageISO3>ENG</LanguageISO3>
                                <Address>
                                    <Building>
                                        <string>CHENG SAN CENTRE</string>
                                    </Building>
                                    <SubBuilding>
                                        <string>FLOOR 01</string>
                                        <string>UNIT 2385</string>
                                    </SubBuilding>
                                    <Street>
                                        <string>ANG MO KIO AVENUE 10</string>
                                    </Street>
                                    <HouseNumber>
                                        <string>528</string>
                                    </HouseNumber>
                                    <Locality>
                                        <string>SINGAPORE</string>
                                    </Locality>
                                    <PostalCode>
                                        <string>560528</string>
                                    </PostalCode>
                                    <Country>
                                        <string>SINGAPORE</string>
                                    </Country>
                                    <DeliveryAddressLines>
                                        <string>528 ANG MO KIO AVENUE 10</string>
                                        <string>#01-2385 CHENG SAN CENTRE</string>
                                    </DeliveryAddressLines>
```

```
                                    <CountrySpecificLocalityLine>
                                        <string>SINGAPORE 560528</string>
                                    </CountrySpecificLocalityLine>
                                    <FormattedAddress>
                                        <string>528 ANG MO KIO AVENUE 10</string>
                                        <string>#01-2385 CHENG SAN CENTRE</string>
                                        <string>SINGAPORE 560528</string>
                                        <string>SINGAPORE</string>
                                    </FormattedAddress>
                                    <AddressComplete>528 ANG MO KIO AVENUE 10;#01-2385 CHENG
SAN CENTRE;SINGAPORE 560528;SINGAPORE</AddressComplete>
                                </Address>
                            </ResultData>
                        </ResultDataSet>
                    </Result>
                </Results>
            </ProcessResult>
        </ProcessResponse>
    </soap:Body>
</soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Singapore addresses.

The following table lists the types of enrichment that you can add to a Singapore address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Singapore addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Singapore addresses.

Frequently Asked Questions

Can Informatica Address Verification return geocodes for a Singapore address when the post code is absent?

Yes, Address Verification can return interpolated geocodes to house number-level accuracy if the post code is absent and the other elements are present. Address Verification returns a geocoding status code of EGC8 in such cases.

How does Address Verification handle geocodes for a Singapore address when the address contains the street name only?

When the address contains the street name only, Address Verification provides geocodes at locality level and returns a geocoding status code of EGC6.

How does Informatica Address Verification handle a Singapore address that contains only the post code?

In Singapore, a post code can identify a delivery point. If the post code identifies a single address, batch processing will verify the address and return the complete address information. In such cases, Address Verification can return interpolated geocodes to house number-level and a geocoding status code of EGC8.

Note: In some cases, a Singapore post code identifies a property with multiple possible addresses, such as a school complex. In such cases, you can submit the post code in fast completion mode to find the corresponding addresses. When you select an address from the fast completion suggestions, Address Verification can return interpolated geocodes to house number-level with a geocoding status code of EGC8.

Reference Address Databases

Informatica updates the Singapore reference address databases quarterly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Singapore addresses:

- `SGP5BI.MD`. Batch and interactive modes.
- `SGP5CA.MD`. CAMEO data.
- `SGP5FC.MD`. Fast completion mode.
- `SGP5GC.MD`. Standard geocoding.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud®.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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